

Tony RuiKang OuYang

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EDUCATION

University of Cambridge

DPhil in Advanced Machine Learning

Cambridge, UK
Oct 2025 – June 2028

- Supervisor [Prof. José Miguel Hernández-Lobato](#) 🌐
- **Fully-funded** by EPSRC DLA Studentship Award

University of Cambridge

MPhil in Machine Learning and Machine Intelligence

Cambridge, UK
Sep 2023 – Aug 2024

- Graduate with **Distinction (Grade: 77%, Top 20%)**
- Course Advisor & Thesis Supervisor: [Prof. José Miguel Hernández-Lobato](#) 🌐
- Thesis: [Energy-based Neural Sampler for Boltzmann Distribution](#) (Grade: 77.5%)

University of Oxford

Visiting Student in Mathematics and Statistics

Oxford, UK
Jan 2022 – Jul 2022

- Supervisor: [Prof. Gesine Reinert](#) 🌐 and [Prof. Geoff Nicholls](#) 🌐
- Fully funded by Harbin Institute of Technology, Shenzhen

Harbin Institute of Technology, shenzhen

BEng in Data Science

Shenzhen, CN
Sep 2019 – Jun 2023

- Graduate with **Distinction (Grade: 90%, Top 10%)**
- Thesis Supervisor: [Yi Zhao](#) 🌐
- Thesis: [Implicit Local Node Embedding Synchronisation for Graph Autoencoders](#) (Grade: 90%)

EXPERIENCES

MolSS reading group

Lead Organiser

London, UK
May. 2025 - present

- A [community](#) 🌐 for machine learning in molecular science

MLG@Cambridge

Research Assistant in Probabilistic Machine Learning

Cambridge, UK
Oct 2024 – Oct 2025

Supervisor: [Prof. José Miguel Hernández-Lobato](#) 🌐

The Alan Turing Institute

Research Assistant

London, UK
Apr. 2022 - Jul. 2022

Supervisor: [Prof. Gesine Reinert](#) 🌐

- Research about scalable Graph Neural Network and Federated Learning for data privacy

PUBLICATIONS

*Equal Contributions

- **OuYang, RuiKang***, Grenioux, Louis*, and Hernández-Lobato, José Miguel (2026). [A Diffusive Classification Loss for Learning Energy-based Generative Models](#). *ICML*.
- Yu, James Boran, **OuYang, RuiKang**, Horwood, Julien, and Hernández-Lobato, José Miguel (2026). [Stochastic Interpolants in Hilbert Spaces](#). *Preprint*.
- **OuYang, RuiKang***, Rissanen, Severi*, He, Jiajun, Chen, Wenlin, Heinonen, Markus, Solin, Arno, and Hernández-Lobato, José Miguel (2025). [Progressive Tempering Sampler with Diffusion](#). *ICML*.
- He, Jiajun*, Du, Yuanqi*, Vargas, Francisco, Zhang, Dinghui, Padhy, Shreyas, **OuYang, RuiKang**, Gomes, Carla, and Hernández-Lobato, José Miguel (2025). [No Trick, No Treat: Pursuits and Challenges Towards Simulation-free Training of Neural Samplers](#). *FPI@ICLR*.
- **OuYang, RuiKang***, Qiang, Bo*, and Hernández-Lobato, José Miguel (2025). [BNEM: A Boltzmann Sampler Based on Bootstrapped Noised Energy Matching](#). *TMLR*.
- **OuYang, RuiKang** (2024). [Energy-Based Diffusion Neural Sampler for Boltzmann Densities](#). *MPhil Thesis*.

- **OuYang, RuiKang**, Elliott, Andrew, Limnios, Stratis, Cucuringu, Mihai, and Reinert, Gesine (2023). [L2G2G: A Scalable Local-to-Global Network Embedding with Graph Autoencoders](#). *CNA*.

ACADEMIC SERVICES

Reviewing

- Conference: ICLR 2026
- Workshop: DELTA@ICLR-2025, FPI@NeurIPS-2025, SPIGM@NeurIPS-2025, DynaFront@NeurIPS-2025

SUPERVISION AND TEACHING

MEng Project Supervision

Sunaabh Trivedi	<i>Better Diffusion Models for Molecular Modelling</i>	University of Cambridge
		<i>Oct 2024 - June 2025</i>
Luran Wang	<i>Towards better Neural Samplers for Boltzmann Distribution</i>	University of Cambridge
		<i>Oct 2024 - June 2025</i>

MPhil Thesis Supervision

Antonio Franca	<i>Transferable Neural Transports for Parallel Tempering</i>	University of Cambridge
		<i>Apr 2025 - Aug 2025</i>
James Yu	<i>Stochastic Interpolants in Hilbert Spaces</i>	University of Cambridge
		<i>Apr 2025 - Aug 2025</i>

AWARDS AND ACHIEVEMENTS

Distinction in MPhil Degree, University of Cambridge	<i>2024</i>
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Ranked top 20% in MPhil in Machine Learning and Machine Intelligence

Travel and Research Grants, Wolfson College, University of Cambridge	<i>2023</i>
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Award for supporting my travelling for conference

Outstanding Graduates, Harbin Institute of Technology (Shenzhen)	<i>2023</i>
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Ranked top 10% in HITsz

Distinction in BEng Degree, Harbin Institute of Technology (Shenzhen)	<i>2023</i>
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Ranked top 10% in BEng in Data Science

International Exchange Program Scholarship, HITsz	<i>2022</i>
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Scholarship for visiting in Oxford, valued at about 200,000 CNY

OTHERS

Athletic

Player, Wolfson College (University of Cambridge) Football Varsity	<i>2023 - 2024</i>
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Player, Harbin Institute of Technology (Shenzhen) Football Varsity	<i>2020 - 2023</i>
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Captain, Harbin Institute of Technology (Shenzhen) Science School Football Team	<i>2021 - 2023</i>
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2nd place of HITsz Football League Season 22-23	<i>2023</i>
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2nd place of HITsz Football League Season 20-21	<i>2021</i>
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2nd place of HITsz Futsal League Season 20-21	<i>2021</i>
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3rd place of HITsz Football League Season 19-20	<i>2020</i>
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Languages

- Cantonese (1st Native)
- Chinese Mandarin (2nd Native)
- English (Professional working proficiency; IELTS 7.0)
- French (Elementary proficiency)